# 📏 Measuring Tools: Reading

Small Engines Overhaul · Unit 1 · Level 3 (closer to the class work) · goes with the Measurement Lab, Micrometer Handout and cylinder bore video

Name: _____

## Precision Measuring in the Engine Shop

Engine parts are made to very exact sizes. Mechanics measure them to one thousandth of an inch, written 0.001. A sheet of paper is about three thousandths thick, so these tools measure very small differences.

A dial caliper has jaws that slide along a bar. The outside jaws measure the outside of a part, the inside jaws measure a hole, and the depth rod measures how deep a hole is. To read it, read the inches and tenths on the bar, then add the thousandths on the dial.

A micrometer is even more exact. You turn the thimble until the spindle touches the part, using the ratchet so you do not squeeze too hard. Read the sleeve (each line is 0.025) and add the thimble (each line is 0.001).

A feeler gauge is a set of thin blades with known thicknesses. It measures small gaps like spark plug gap, ring end gap and valve clearance. The repair manual gives the specification for every part. If your measurement is outside the spec, the part is worn and may need to be replaced.

## Check your reading

1. What is a thousandth of an inch, written as a decimal?

2. How do you read a dial caliper?

3. What does it mean if a part measures outside the spec?